

Power Tips: How to Find Switching Loops in a Power Supply



Robert Taylor

Power-supply layout is one of the most important parts of being able to create a successful power design. Everyone has an opinion on how to do it and reasons why their way is the best way. The truth is that many different solutions will work; unless you really make a mess of the layout, the supply will probably function.

Of course, there are some universal rules, like:

- Don't run sensitive signals under fast-switching signals. In other words, don't run the feedback trace under the switch node.
- Make sure that the power-carrying traces and planes are large enough to support the current.
- Try to keep at least one continuous ground plane.
- Use enough vias (typically 1A per via is a good place to start) to connect planes together.

Beyond these basic layout rules, I always start first by identifying the switching loops and then identifying which ones will have a high-frequency switching current. Figure 1 shows an example of a simplified power stage for a buck power supply (schematic and layout).

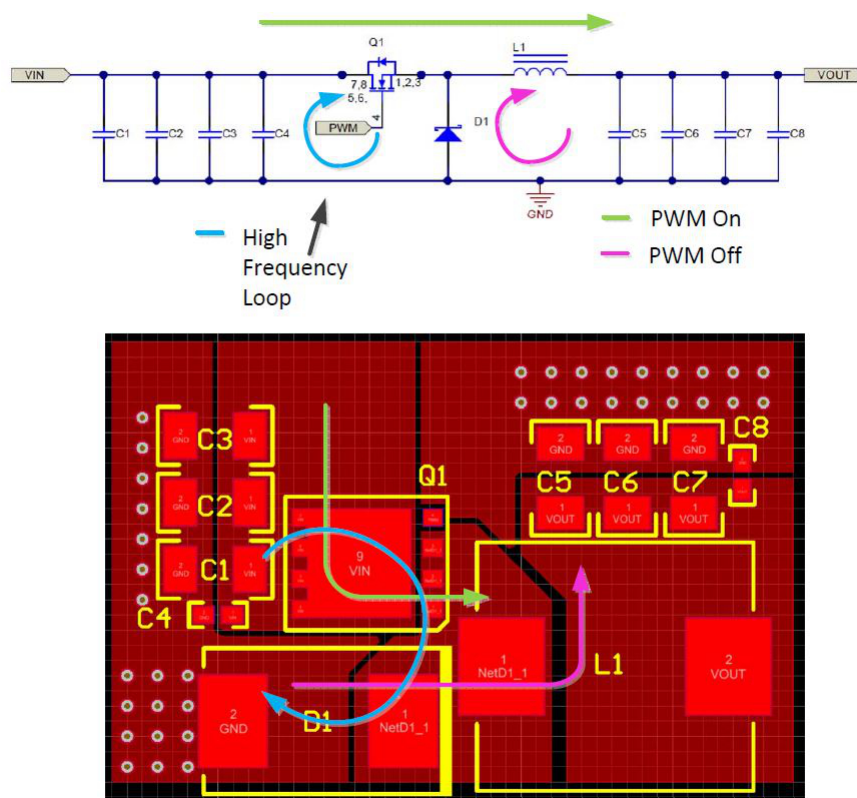


Figure 1. Buck Power-supply Schematic and Layout

In a buck power supply, there are two states (assuming continuous conduction mode): when the control switch (Q1) is on and when the control switch is off. When the control switch is on, current flows from the input to the

inductor. When the control switch is off, current continues to flow in the inductor and through the diode (D1). The current is continuous to the output.

There is pulsing current on the input, however, and that's the portion of the layout you need to pay attention to. In [Figure 1](#), this loop is labeled "High Frequency Loop" and is shown in blue. Your primary objective of the layout should be to connect the Q1, D1 and input capacitors with the shortest, lowest inductance loop possible. Making this loop small will minimize noise generated by the switching. If you neglect to do this, the supply will not work very well.

The procedure for identifying the switching loops applies to all power-supply topologies. Step by step, the procedure is to:

1. Identify the current paths during the on state.
2. Identify the current paths during the off state.
3. Find where the continuous current exists.
4. Find where the discontinuous current exists.
5. Minimize any loop that has discontinuous current.

This list shows the critical loops for a given power-stage configuration:

- Buck – input capacitor loop.
- Boost – output capacitor loop.
- Inverting buck-boost – input and output capacitor loops.
- Flyback – input and output capacitor loops.
- Fly-Buck™ - input capacitor loop.
- SEPIC – output capacitor loop.
- Zeta – input capacitor loop.
- Forward, half bridge, full bridge – input capacitor loop.

Power-supply layout is an art form. Everyone has their own way of doing it, and most of the time that is OK. Just make sure that when you are looking at placing the parts for a power stage, you identify the high-frequency switching loop; you will save yourself some time and headaches down the road.

Additional Resources

- See my latest EE Times Power Tip post, "[Multiphase Power Supplies: Not Just For High Current](#)" for more about this topic.
- Watch a [video](#) to get layout tips on reducing radiated EMI in a DC/DC buck regulator design.
- Explore more [power-supply topics](#).
- Watch Power Tips [videos](#) to help with your design challenges.

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